MODULE 1 INTRO TO ANATOMY

The Language of Anatomy.

INTRODUCTION TO ANATOMY

The shared vocabulary the rest of the course is built on. Every position, plane, direction, and region named here gets used in lab, on exams, and in every clinical description you will ever read.

COMPETENCIES W1-LEVELS-ORGANIZATION, W1-ANATOMICAL-POSITION, W1-PLANES-SECTIONS, W1-DIRECTIONAL-TERMS, W1-REGIONAL-TERMS

BY THE END

1. Define anatomy and physiology and name the branches of each.
2. List the six levels of structural organization in order and give an example of each.
3. Name the eleven organ systems and the six basic life processes.
4. Place the body in anatomical position and explain why it is the reference for every directional term.
5. Name the planes and sections and use the paired directional terms precisely.
6. Identify the major anterior and posterior regional terms.
7. Separate disease from disorder and signs from symptoms.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Week 1

drsrennie-stack.github.io/new-build-bio4-solano/week1-concept-videos.html

Anatomy and Physiology

Anatomy is the study of the structures in the body. **Physiology** is the study of how those structures function, how they work. The two are inseparable in practice. Structure sets the limits on what a part can do, and function explains why a structure is shaped the way it is. This course is the anatomy half.

Branches of anatomy

- **Gross (macroscopic) anatomy**, structures visible to the unaided eye. Studied **regionally** (all structures in one body area), **systemically** (one organ system at a time), or as **surface anatomy** (internal structures located by external landmarks).
- **Microscopic anatomy**, structures too small to see unaided. Includes **cytology** (cells) and **histology** (tissues).
- **Developmental anatomy**, structural change across the lifespan. **Embryology** is the subdivision covering development before birth.
- **Pathological anatomy**, structural change caused by disease.
- **Radiographic anatomy**, internal structure seen through imaging.

Branches of physiology

- **Neurophysiology**, function of the nervous system.
- **Endocrinology**, hormones and how they regulate function.
- **Cardiovascular physiology**, function of the heart and vessels.
- **Respiratory physiology**, function of the airways and lungs.

- **Renal physiology**, function of the kidneys.
- **Immunology**, how the body defends itself.
- **Pathophysiology**, functional change caused by disease.

Levels of Structural Organization

The body is organized from the simplest level to the most complex. Each level is built from the one before it, so a failure at a lower level always has consequences above it.

THE SIX LEVELS, LEAST TO MOST COMPLEX

Level	What it is	Example
Chemical	Atoms and molecules, the smallest level	Oxygen, water, a protein
Cellular	The smallest living unit of the body	A keratinocyte, a neuron
Tissue	Groups of similar cells performing a specific function	Stratified squamous epithelium
Organ	A structure composed of two or more tissue types with a defined job	The skin, the stomach
Organ system	Organs working together toward one broad function	The integumentary system
Organismal	All organ systems together, the entire living being	The person

The Eleven Organ Systems

SYSTEM, MAJOR COMPONENTS, PRIMARY FUNCTION

System	Major components	Primary function
Integumentary	Skin, hair, nails, cutaneous glands	Covers and protects, regulates temperature, senses the environment
Skeletal	Bones, cartilage, ligaments, joints	Supports, protects, stores minerals, houses blood formation
Muscular	Skeletal muscles and their tendons	Produces movement, maintains posture, generates heat
Nervous	Brain, spinal cord, nerves, sensory receptors	Fast control, detects and responds to change
Endocrine	Pituitary, thyroid, adrenals, pancreas, gonads	Slow control through hormones
Cardiovascular	Heart, blood vessels, blood	Transports oxygen, nutrients, wastes, and heat
Lymphatic and immune	Lymph vessels and nodes, spleen, thymus, tonsils	Returns fluid, filters lymph, mounts immune defense
Respiratory	Nasal cavity, pharynx, larynx, trachea, bronchi, lungs	Gas exchange between air and blood
Digestive	Alimentary canal plus liver, gallbladder, pancreas	Breaks down food, absorbs nutrients, eliminates residue
Urinary	Kidneys, ureters, bladder, urethra	Filters blood, eliminates nitrogenous waste, balances water and electrolytes
Reproductive	Gonads, ducts, accessory glands, external genitalia	Produces gametes and offspring

The Six Basic Life Processes

These distinguish living things from non-living things. Every one of them shows up again in a later unit.

1. **Metabolism**, the sum of all chemical reactions in the body. **Catabolism** breaks molecules down, **anabolism** builds them up.
2. **Responsiveness**, the ability to detect and react to a change in the internal or external environment.
3. **Movement**, motion of the whole body, of individual organs, of single cells, or of structures inside a cell.
4. **Growth**, an increase in size, from more cells, larger cells, or more material between cells.
5. **Differentiation**, the change of an unspecialized cell into a specialized one. Stem cells become keratinocytes, neurons, muscle fibers.
6. **Reproduction**, either the formation of new cells for growth and repair, or the production of a new individual.

Anatomical Position

Anatomical position is the standard reference posture: the body erect, feet flat and slightly apart, head and eyes facing forward, arms at the sides, palms facing forward.

Why it matters. Every directional term assumes this position. That way a description does not change with how the person is actually standing, sitting, or lying. Right and left are always the patient's right and left, not yours.

Two related terms describe a body lying down. **Supine** is lying face up, on the back. **Prone** is lying face down, on the front.

Body Planes and Sections

A **plane** is an imaginary flat surface passing through the body. The **section** is the cut surface that plane produces. The **midline** is the imaginary vertical line running from the top of the head, through the nose, down the spine, to the pelvis. It divides the body into equal left and right halves and serves as the reference point for describing locations relative to the body's center.

PLANES AND WHAT EACH ONE DIVIDES THE BODY INTO

Plane	Orientation	Divides the body into
Sagittal	Vertical	Left and right parts
Midsagittal (median)	Vertical, on the midline	Equal left and right halves
Parasagittal	Vertical, off the midline	Unequal left and right parts
Frontal (coronal)	Vertical	Anterior and posterior portions
Transverse (horizontal)	Horizontal	Superior and inferior portions, a cross section
Oblique	At an angle other than 90 degrees	Angled portions, between horizontal and vertical

Directional Terms

Directional terms come in pairs of opposites, and they are always given from the patient's perspective in anatomical position. A structure lying between a medial and a lateral structure is called **intermediate**.

PAIRED DIRECTIONAL TERMS

Term	Its opposite	The relationship the pair describes
Superior	Inferior	Toward the head or above, versus away from the head or below
Anterior (ventral)	Posterior (dorsal)	Toward the front, versus toward the back of the body
Medial	Lateral	Toward the midline, versus away from the midline
Proximal	Distal	Closer to the point of attachment to the trunk, versus farther from it. Limbs only
Superficial	Deep	Toward the body surface, versus away from the surface
Central	Peripheral	Near the center of a structure, versus away from its center
Ipsilateral	Contralateral	On the same side of the body, versus on the opposite side

Regional Terms

The body divides into an **axial** region, the head, neck, and trunk, and an **appendicular** region, the limbs. Each named region has a precise term. Learn these as a location on your own body, not as a word list.

Anterior view

ANTERIOR REGIONAL TERMS

Term	Region	Term	Region
Frontal	Forehead	**Inguinal**	Groin
Cranial	Skull or head	**Pubic**	Area near the genital region
Facial	Face	**Brachial**	Upper arm
Orbital or ocular	Eye region	**Antebrachial**	Forearm
Buccal	Cheek	**Antecubital**	Front of the elbow
Otic	Ear region	**Carpal**	Wrist
Nasal	Nose	**Pollex**	Thumb
Oral	Mouth region	**Palmar**	Palm of the hand
Mental	Chin	**Digital or phalangeal**	Fingers or toes
Cervical	Neck region	**Patellar**	Kneecap
Thoracic	Chest area	**Crural**	Leg, the shin area
Mammary	Breast region	**Tarsal**	Ankle
Abdominal	Below the chest and above the pelvis	**Pedal**	Foot
Umbilical	Navel or belly button	**Hallux**	Big toe
Pelvic	Lower abdomen or hip area		

Posterior view

POSTERIOR REGIONAL TERMS

Term	Region	Term	Region
Cephalic	Head	**Sacral**	Area between the hips
Cervical	Neck	**Gluteal**	Buttock region
Acromial	Point of the shoulder	**Femoral**	Thigh
Dorsal	Back	**Popliteal**	Back of the knee
Olecranal	Back of the elbow	**Sural**	Calf
Lumbar	Lower back or loin area	**Calcaneal**	Heel
Plantar	Sole of the foot		

Clinical Vocabulary

Disease versus disorder

Disease is a pathological condition caused by identifiable physical or biochemical changes, often from infection, genetics, or environmental factors. A disease typically has a clear set of symptoms, a predictable progression, and an identifiable cause. Examples: diabetes mellitus, a metabolic disease involving insulin production or utilization. Tuberculosis, an infectious disease caused by **Mycobacterium tuberculosis**.

Disorder is a disruption of normal physical or mental function that often lacks a single clear cause. Disorders may result from complex interactions among genetic, environmental, and lifestyle factors. Examples: anxiety disorder, marked by excessive fear or worry. Functional bowel disorders such as irritable bowel syndrome, which have no clear structural abnormality.

TWO MORE WAYS TO SORT A CONDITION

Term	Meaning	Example
Local	Affects one part or one limited region of the body	A skin abscess
Systemic	Affects the whole body or several organ systems at once	Sepsis, influenza
Acute	Sudden onset, short duration	Appendicitis
Chronic	Gradual onset, long duration, often lifelong	Type 2 diabetes mellitus

Signs versus symptoms

A **sign** is objective. It is a measurable physical manifestation that a healthcare professional can observe. Fever measured as an elevated temperature, a visible rash, an increased heart rate on a pulse check or ECG.

A **symptom** is subjective. It is the patient's own experience of illness, and it cannot be measured directly by an observer. Pain, fatigue, nausea.

THE SAME CONDITION, SORTED BOTH WAYS

Condition	Signs (observed)	Symptoms (reported)
Influenza	Fever, nasal discharge, coughing	Fatigue, body aches, sore throat
Anxiety disorder	Increased heart rate, sweating during episodes	Feeling of dread, difficulty concentrating

What goes into a clinical diagnosis

- **History**, what the patient reports. Symptoms, timing, what makes it better or worse, past conditions, family history.
- **Physical examination**, the signs the clinician can observe, palpate, percuss, or listen for.
- **Vital signs**, temperature, pulse, respiratory rate, blood pressure, oxygen saturation.
- **Laboratory testing**, blood, urine, culture, tissue.
- **Imaging**, radiograph, CT, MRI, ultrasound. The plane you image in determines what you can see.
- **Differential diagnosis**, the ranked list of conditions that could explain the findings, narrowed as evidence comes in.

WHY THE WORDS ARE NOT OPTIONAL

Anatomical terms remove ambiguity. A chart that reads "wound on the anterior, distal forearm" means the same thing to every clinician on every shift, no matter how the patient is positioned. Imaging is built on the planes: a CT slice is a transverse section, and an MRI is often read in the sagittal and frontal planes, so knowing the plane tells you which view you are looking at. Paired terms have to be used precisely. Proximal and distal describe the limbs only, never the trunk. Ipsilateral and contralateral matter in neurology, where a lesion on one side of the brain can show its effect on the opposite side of the body.